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Gadget models of cod-capelin-shrimp interactions in Icelandic waters

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Abstract

In Icelandic waters there are known to be strong links between cod and capelin and between cod and shrimp, with the condition of cod related to capelin abundance and shrimp abundance affected by predation by cod. The development of a model to assess the importance of capelin abundance to cod, and the corresponding impact of the capelin fishery on cod is therefore of some importance. Similarly, in assessing the abundance of shrimp, understanding the impact of cod predation might be expected to improve both the understanding of population dynamics and abundance estimates. Modelling these scenarios requires multispecies, multiarea, multifleet models. In both cases the spatial overlap of the predator and prey needs to be taken into account. There are also large temporal changes in the availability of capelin as prey as they are only present on the Icelandic shelf for part of the year. Each of these species has different modelling requirements and differing levels of data availability. The development of models to address these issues is described, with levels of model complexity compared and evaluated.

Keywords: Gadget, multispecies, statistical model, cod, capelin, shrimp, migration

1 Introduction

The marine ecosystem off Iceland contains several important players. Considered in isolation, the cod (*Gadus morhua*) stock is an important demersal species whose population dynamics have been studied extensively (e.g. Stefansson (1992) and Schopka (1994)). An interesting twist on the population dynamics is cannibalism, which appears to exist, though not of a dominating importance (Bogstad et al. 1994). The cannibalism has been taken into account when evaluating utilisation strategies for the cod stock (Baldursson et al. 1996).

The capelin (*Mallotus villosus*) is a very abundant (though also highly variable) prey species and has also been studied extensively (Vilhjalmsson 1994). When considering the part of the ecosystem related to commercial species, the most studied link is between cod and capelin, where capelin is an important food for cod (Magnusson and Palsson 1989; Magnusson and Palsson 1991a). Both of these finfish species sustain fisheries of considerable commercial importance (MRI 2004).

Shrimp (*Pandalus borealis*) is also a commercially important species, and another important link is between cod and shrimp, where predation by cod forms a considerable part of the mortality

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on the shrimp (Magnusson and Palsson 1991b). This effect has been captured using models of interactions as a part of routine assessments of shrimp (Stefansson et al. 1994; MRI 2004).

The various links between these three species have been studied using simple regression analyses (Stefansson et al. 1998b). The analyses support these conclusions but are difficult to convert into absolute numbers, though not impossible (Steinarsson and Stefansson 1996). These links have been used to develop harvest control rules for cod (Danielsson et al. 1997), as have sequences of more complex links (Stefansson et al. 1998a).

Models of marine ecosystems tend to be simplified in different ways depending on the purpose of the analysis conducted. As noted above, a very simple model can be in the form of a simple linear regression, possibly taking into account the statistical properties of the data. A very detailed model of the predation process could be a strictly mathematical model with no recourse to data fitting. Many model types fall somewhere in between these extremes, taking into account a varying level of detail in the mathematical model of the biological process and the statistical description of the data to which the model is fitted.

Although the simple models should be expected to illustrate the most important relationships, several problem issues arise. These include variable species overlap, seasonal and stage-dependent migratory behaviour as well as competing mortality factors which can only be sensibly considered in joint analyses.

The models to be developed in this paper are therefore implemented using the scheme set out in Stefansson and Palsson (1998) and implemented using a computer program, Gadget, described in detail by Begley and Howell (2004) and Begley (2004). These models are statistical in nature, i.e. they are based on fitting to data using likelihood functions. They are also parametric in that a prediction is based on a parameter value alone, ideally using data only through the formal data fitting procedure. Finally, these models can take into account spatial and temporal variation in the species under consideration.

When developing these models it has proved important to first develop single-species models of fairly high quality for each species in question. Given the different data availability for e.g. cod and shrimp, these models become very different in nature. Thus, although the models may be internally age-disaggregated, there are no true age data for shrimp. Similarly, although most traditional assessments of cod in the North Atlantic use catches in numbers, disaggregated by age, such data are highly processed and the present paper uses data closer to the original samples, in accordance with common statistical procedure.

Earlier versions of such models have been used to analyse the cod-capelin interactions (Bjornsson 1998), based on Bormicon (Stefansson and Palsson 1997). Most of the earlier work has relied heavily on unproven and hardwired assumptions and weights given to different data sources. Given the problems in such approaches (Stefansson 1998), the emphasis in the present paper is on the cod-shrimp interaction, using appropriate statistical estimation methods at each stage of model development.

2 Model definition

The fishery for the offshore shrimp stock takes place mainly in the Gadget divisions 103, 104, 105 and 111 (figure 1). The definition of these divisions is described in Taylor (2003b). The cod stock, however, is predominantly found on the Icelandic shelf (< 500m). The simplest spatial resolution for a cod-shrimp stock model is to develop a two area model with a northern area incorporating divisions 103, 104 and 105 and a southern area of 101, 102, 107 and 108. Within the model, shrimp only exist on the northern area, whereas cod are on both and migrate between the two areas.

Given the spatial structure of the final cod-shrimp model, two preliminary single species models can be developed independently. This enables the impact of cod predation on shrimp to be assessed, and given the difficulties of estimating migration, it is more efficient to work with a simpler model in the first instance.

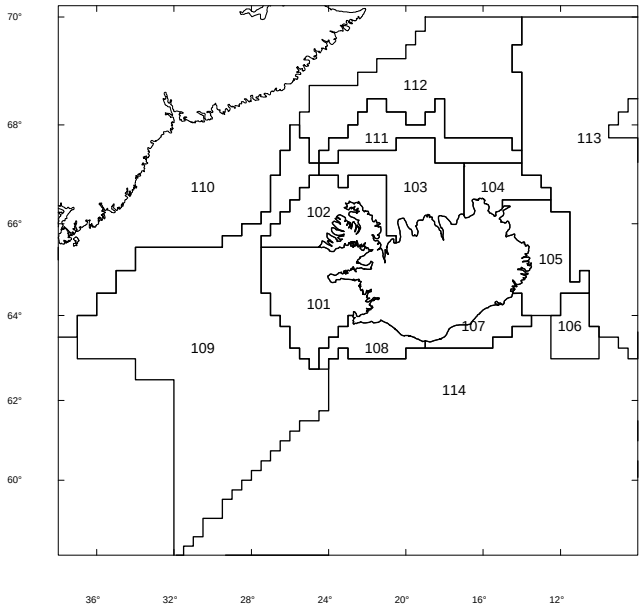


Figure 1: Map showing some of the datawarehouse divisions for the area around Iceland.

One of the aims of the cod model is to estimate migration of cod between the areas using the methodology described in Hannesson et al. (2004). A simpler alternative would be to fix the migration rates to values estimated externally to Gadget, e.g. using the method described for capelin by Dereksdottir et al. (2003).

The interest in linking cod and shrimp is in the impact of predation on shrimp rather than the effect of consumption on cod. As shrimp is not a major part of the diet of cod (Magnusson and Palsson 1991b), consumption of shrimp is not considered to have a significant impact on cod growth (Bjornsson 1998) and in this implementation cod growth is independent of consumption.

The approach to parameter estimation, through iterative weighting of each likelihood component is similar to that described in Taylor et al. (2004) and will not be discussed here.

Input data and likelihood component data were extracted from the datawarehouse described in Kupca and Sandbeck (2003) with the original data sources and aggregation methods described in Taylor (2003a). In all cases, except tagging data, the likelihood functions are those used for the cod models in Taylor et al. (2004).

3 Shrimp model

3.1 Data and likelihood functions

Age data are only available for shrimp by using methods to disaggregate length frequency data (e.g MacDonald and Pitcher (1979)). Age disaggregation is problematic, especially for the older ages as the modes in the length distribution are difficult to identify. As can be seen in figure 2, only the modes of the first three cohorts are easily distinguished. For this reason, age data are only used in this model to initialise the population and to determine the growth parameters which are then fixed.

Data used in the shrimp model are:

Input data:

- Landings data obtained from the Directorate of Fisheries data base.

Biological sampling likelihood data:

- Length distribution, aggregated on 1mm intervals, from sampling of commercial catches by month from 1988 to 2002.
- Length distribution, aggregated on 1mm intervals, from the main net of the offshore shrimp survey from 1988 to 2002 (Skuladottir et al. 2000).

Survey index likelihood data:

- Four survey indices from the main net of the offshore shrimp survey (figures 2 & 3) with one datum for each group for the years 1988–2002.
- Two survey indices from the juvenile bag of the offshore shrimp survey (figures 2 & 3) with one datum for each group for the years 1988–2002.

Four indices are calculated for the main net (and two for the juvenile bag) by splitting the length distribution into groups, with the division based on the 'typical' structure of the length distribution for the survey over all years. The first three length groups are considered to be cohorts. The division of the length distributions into groups is illustrated for two years in figure 2. The length group divisions are identical for the two data sources and are derived from length distributions on 0.5mm intervals. The intervals in cm are: (0.60,1.10), [1.10,1.65), [1.65,2.15) and [2.15,3.20) for groups 1–4. This provides two survey indices for the younger shrimp, with likelihood components that can be weighted simultaneously in the iterative weighting scheme. The appropriateness of these divisions as indicators of cohorts varies between years but as the indices are identified as length groups rather than age groups and the two sources of survey indices are collected simultaneously this is not a problem.

For the shrimp surveys, the slope and intercept of the log-linear regression between the survey and modelled data are estimated for length groups 1 and 2 with the slope fixed to 1 for the third and fourth length groups.

In figures 3 & 4 it can be seen that for length groups 1 and 2, the trends from the juvenile bag and main trawl net are similar for length group 1 (correlations between nets of 0.687 and 0.540 for length groups 1 and 2 respectively) although the time series do not entirely agree. The selection

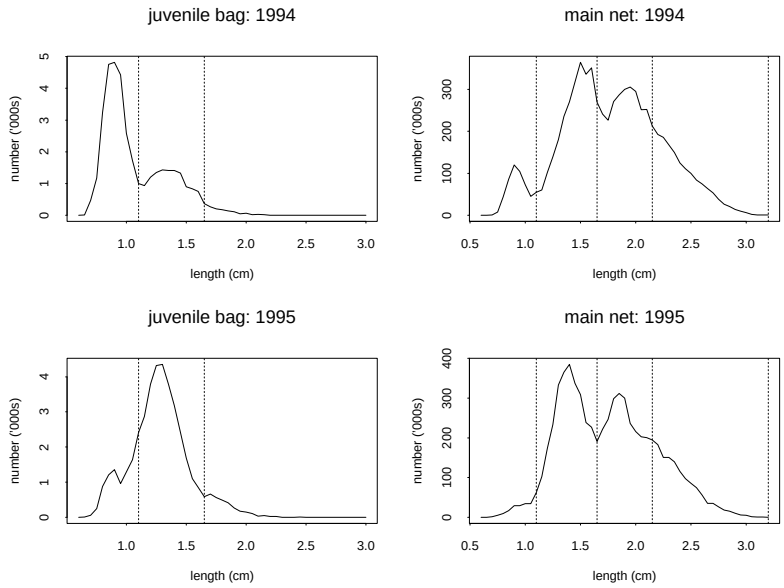


Figure 2: Length distributions from 1994 and 1995 for the juvenile bag and main net of the offshore shrimp survey. The vertical lines indicate the division into length groups for the survey index likelihood components.

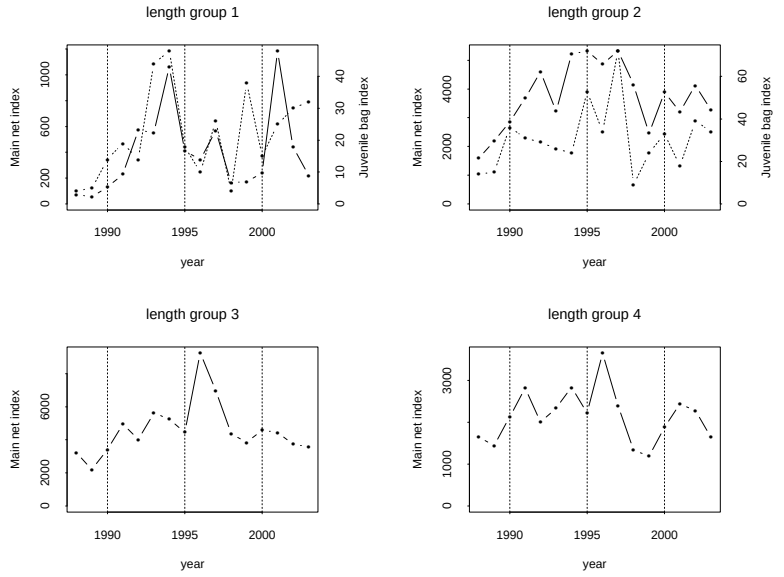


Figure 3: Survey indices for the offshore shrimp survey for both nets by length group. For length groups 1 and 2, the y axis for the main net is on the left and for the juvenile bag on the right with the index for the main net represented by the solid line and the juvenile bag by the dashed line.

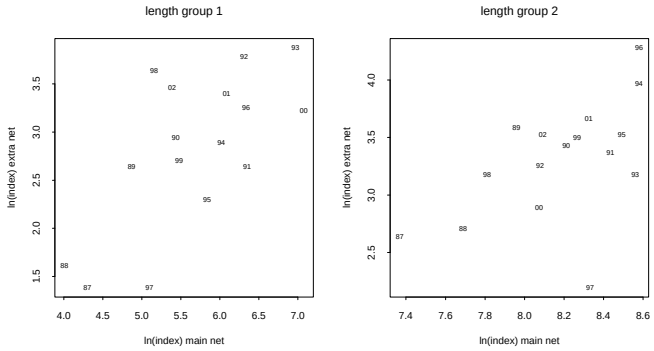


Figure 4: Scatter plots of the (ln transformed) offshore shrimp survey abundance indices. On the left, length group 1 from the main net is plotted with length group 1 from the juvenile bag and on the right, the equivalent plot for length group 2. Each point is labelled with the cohort, assuming group 1 is age 1.

of shrimp in the two nets is different, with the smaller meshed juvenile bag expected to be a more reliable indicator of length group 1 and the main net for length group 2.

Comparing length groups within nets by cohort (figure 5), the relationship between length groups 1 and 2 is considerably stronger in the main net than the juvenile bag (correlations of 0.645 and 0.140 respectively). There is also a strong correlation (0.667) between length groups 2 and 3 of the main net and between length group 1 of the juvenile bag and length group 2 of the main net (correlation of 0.732).

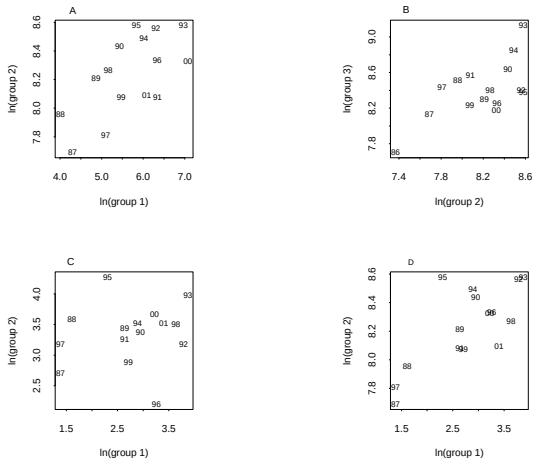


Figure 5: Scatter plots by cohort of the (ln transformed) offshore shrimp survey abundance indices. A) length group 1 vs length group 2 from the main net, B) length group 2 vs length group 3 from the main net C) length group 1 vs length group 2 from the juvenile bag and D) length group 1 from the juvenile bag vs length group 2 from the main net. Each point is labelled with the cohort, assuming group 1 is age 1.

3.2 Model Implementation

The model runs from 1988 to 2002, with monthly timesteps, on one area with two shrimp components – male and female. The males are aged 1 to 6 and the females 3 to 8, with the final age a plus group. Males become female through a maturation function and any not female by the end of their sixth year automatically move into the female stock component. The female component is not split further into immature and mature. A single commercial shrimp fleet operates along with a shrimp survey.

Variable natural mortality has been identified as an important control on the population dynamics of shrimp (Fu and Quinn 2000) and in Icelandic waters predation by cod has been estimated to be high in relation to the catch of the shrimp fishery (Magnusson and Palsson 1991b). As the aim of developing the shrimp model is however to link it with a cod model to investigate the importance of cod predation on shrimp population dynamics, in this implementation natural mortality is constant for all years.

- Length growth is according to the von Bertalanffy function as described in Taylor et al. (2004) with k and L_∞ fixed to 0.17 and 3.501 for both the male and female stock components. The von Bertalanffy parameters were calculated from data (Skuladottir pers. com.). The betabinomial parameter β is fixed at 20, with maximum length group growth $n = 10$ and length intervals of 0.025cm.
- The weight-length relationship is fixed, $W = aL^b$, with different values for the male and female components (Skuladottir pers. com.).
- The transition from male to female is implemented using the maturation function described in Taylor et al. (2004) as a function of length. The parameters defining sex change are estimated from shrimp survey data and fixed.

- Natural mortality at age is fixed to 0.4 for all males, 0.5 for females aged 3 to 6 and 0.6 for age 7 and 8 females. Natural mortality has been set to be higher for females than males as Clark et al. (2000) report that mortality increases sharply after first hatching for mature females for Gulf of Maine shrimp. In some other areas however, natural mortality of males is believed to exceed that of females due to greater predation on smaller shrimp (Fu et al. 2001).
- The mean and standard deviation of length of the initial populations were fixed, and based on a von Bertalanffy curve fitted to data (Skuladottir pers. com.). The weight-length relationships used for weight growth were also used for the initial population.
- The number at age of the initial population (ages 2 to 7) is estimated within the model, with the age distribution estimated and then a multiplier used to scale the population abundance.
- Mean and standard deviation of length of the age 1 recruits are fixed according to the data used to initialise the population.
- The number of age 1 recruits for each year (1988–2002) is estimated within the model.
- To enable the age structure of the initial population to be estimated independently of the survey indices, there is an additional parameter (*mult*) which can be used to scale the abundance of the initial population with the age structure fixed.
- The fleet selection pattern is the same as that used in Taylor et al. (2004) with the length at 50% selected ($L_{50\%}$) and the slope estimated for each fleet.

3.3 Results

Parameter estimation was done in two stages using the weights from table 1:

1. all likelihood data were used,
2. growth, selection (catch and survey) and initial population parameters were fixed, the weights on the length distribution were set to zero and the remaining components were to estimate recruitment and the population abundance.

The higher weight for length groups 1 and 2 is on the juvenile bag and main net respectively (table 1) which is the expected relationship given the survey design. The weights on length groups 3 and 4 of the main net are arbitrary as there are no equivalent data with which to iteratively weight these length groups. A heavy weight on a component containing one datum per year will result in that component fitting perfectly (if optimised for long enough) which means that more than one source of survey index is required for at least some years.

Parameters, fixed and estimated, along with the predicted stock biomass, for a single timestep, and observed catch are shown in figure 6.

Plots of the fitted log-linear regression are shown in figure 7 with the residuals in figure 8. The slopes for length groups 1 and 2 are similar for both data sources, with greater values for the larger meshed main net. As expected the fit of the model to the juvenile bag index is better for length group 1 and to the main net index for length groups 2.

The residual plots for length groups 3 and 4 (figure 8) show that the model overpredicts the group 3 and 4 indices for the final years, while underestimating those for the middle years. This is reflected in the biomass plot in figure 6 where the total stock biomass can be seen to increase from 1998 onwards, whereas the catch decreased in these years. As the first two length groups fit

Component	Weight	sse_f/sse_m
LDs	5479.45	1.209
LDc	1795.12	1.067
I_{1J}	24.74	1.463
I_{2J}	19.18	3.976
I_{1M}	6.04	0.852
I_{2M}	38.05	0.939
I_{3M}	*20.00*	
I_{4M}	*5.00*	

Table 1: Likelihood component weights and sse_f/sse_m, the ratio of the sums of squares from the final model (sse_f) with the minimum sse (sse_m) when that component is heavily weighted. The components are the LD – length distribution, with s and c indicating whether survey or catch data and the survey indices with I_i, $i = 1, 2, 3, 4$ indicating the length group and the J, M subscripts indicate the juvenile bag and main net of the offshore shrimp survey. The weights in italics were not estimated from the iterative procedure.

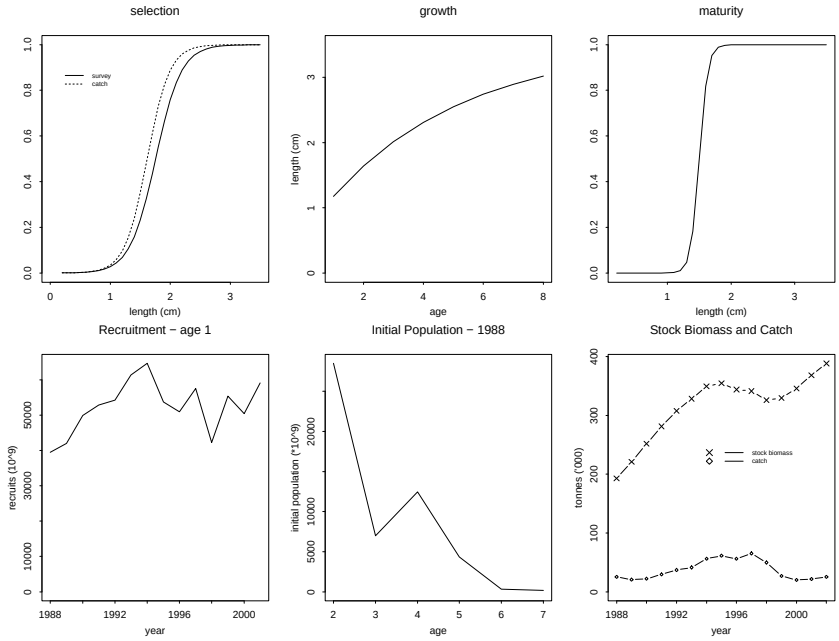


Figure 6: Parameter estimates from the final optimisation: selection patterns of the offshore survey and catch, growth (fixed), sex change (male to female – fixed), number of age 1 recruiting into the model and the initial population ages 2–7. The estimated stock biomass and the observed catch are also shown.

the index data for this time, the problem is entirely due to an over prediction of the abundance of older shrimp. It is possible that this is a result of not taking into account annually varying mortality. With mortality constant for all years and fixed growth and fleet selectivity, recruitment and catch determine the modelled stock structure and abundance. For a stock, such as shrimp, with high predation mortality, natural mortality can be a more important determinant of stock dynamics than recruitment (Fu and Quinn 2000). Another possible cause for the over prediction of older shrimp could be the increased mortality of females after hatching which has been observed for Gulf of Maine shrimp (Clark et al. 2000).

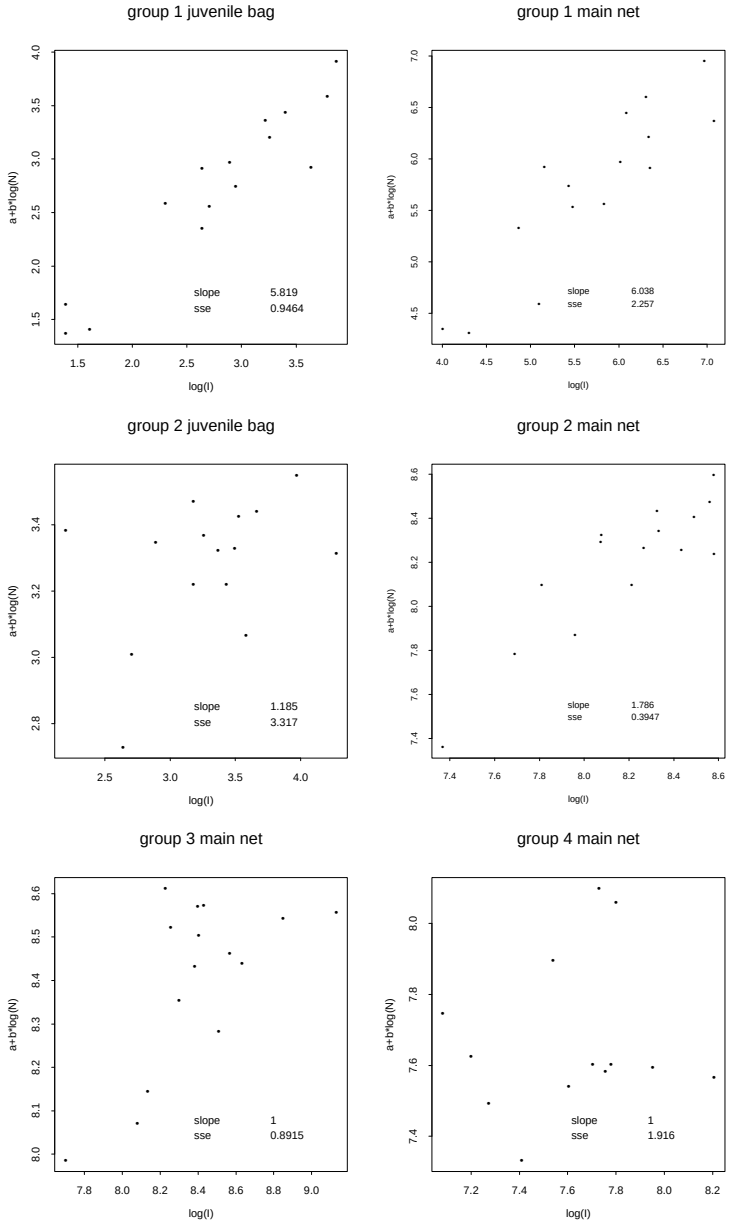


Figure 7: Plots of the log-linear regression as estimated in Gadget, with one plot for each likelihood component and the corresponding slope (estimated for length groups 1 and 2) and sse for each survey index. log(I) is the index and a+b*log(N) the model estimate of the survey index.

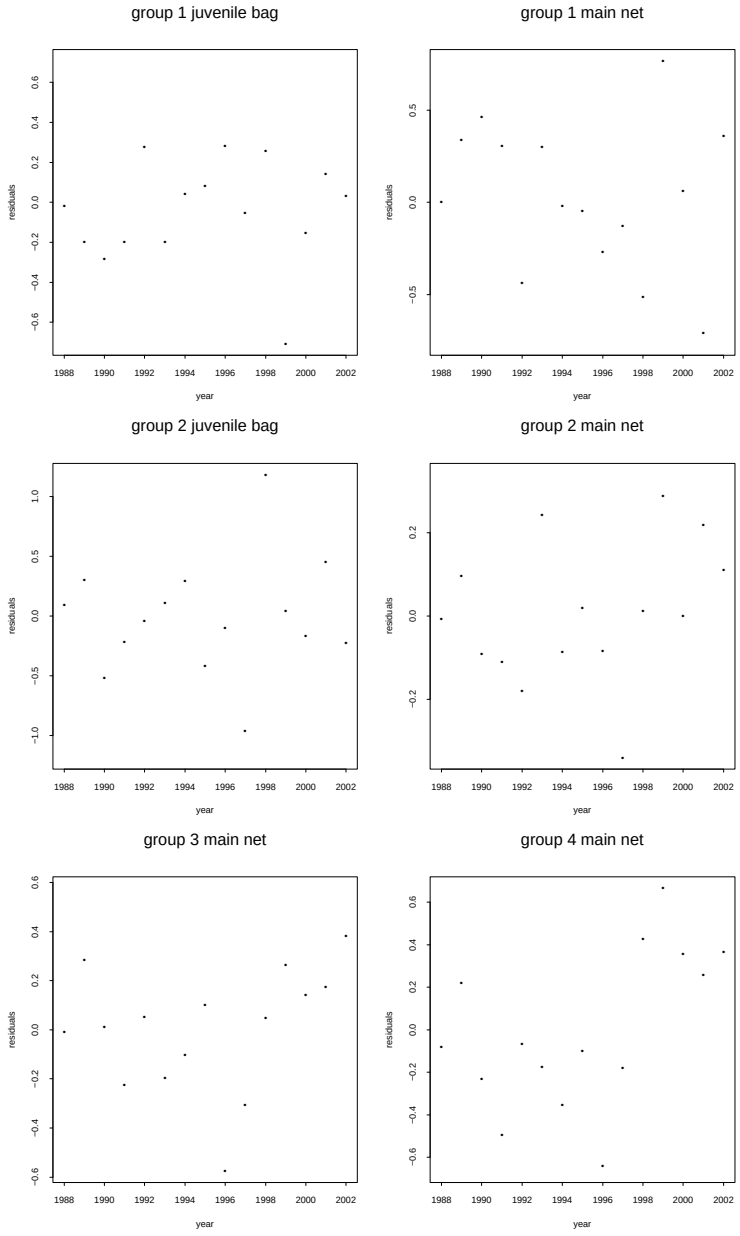


Figure 8: Plots of the residuals from the log-linear regression as estimated in Gadget, with one plot for each likelihood component.

4 2-area cod model

The following description focuses on aspects of the 2-area model which differ from the single area example in Taylor et al. (2004).

4.1 Data and likelihood functions

In a multi-area Gadget model most of the likelihood components can be aggregated over more than one area, area disaggregated within the same likelihood component or considered independent components. Survey indices are an exception to this and must be area disaggregated (although as with all likelihood data do not need to be provided for every area).

Data used in the case study are:

Input data:

- Landings data are obtained from the Directorate of Fisheries data base and are area disaggregated.

Likelihood data:

- Biological sampling likelihood data: length distributions, age-length keys and age distributions for the spring survey and commercial catch as for Taylor et al. (2004) but disaggregated by area.
- Stock composition likelihood data (i.e. proportion mature at length): disaggregated by area.
- Survey index likelihood data: three series of indices for each area for the spring survey (Palsson et al. 1989) for 1985–2002 and the autumn survey (Sigurdsson et al. 1997) for 1995–2002. For each year/area/survey/group one datum is available.
- Four tagging experiments: south 1993, south 1995, south 1996 and north 1993. The tagging likelihood data and components are described in Hannesson et al. (2004) where a more extensive set of tagging likelihood data was used.

The relationship between the spring and autumn surveys can be seen in figure 9 & 10. The correlations between the surveys within the length groups are:

	South	North
group 1	0.759	0.579
group 2	0.654	0.703
group 3	0.659	-0.587

It can be seen that for length groups 1 and 2, the surveys provide similar information for the overlapping years (especially for length group 1 in the south) and that the conflicting information for the third length group is restricted to the northern area where the surveys are negatively correlated. It is also apparent from figures 9 & 11 that the distribution of fish at the time of the surveys is not constant between years. This is particularly true for length group 2 in the spring survey (correlation = 0.35).

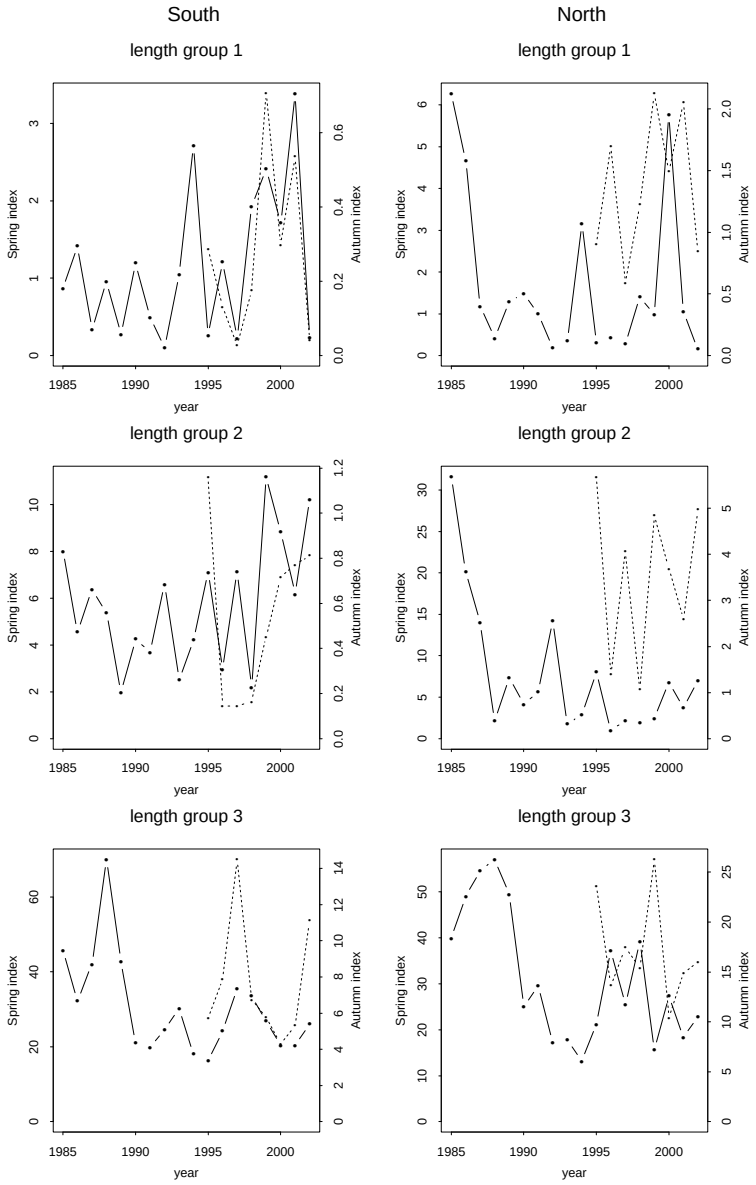


Figure 9: Survey indices for the spring and autumn surveys for each length group, with the y axis for the spring survey on the left and for the autumn survey on the right. Indices for the southern area are on the left in for the northern area on the right. The spring survey is represented by the solid line and the autumn index by the dashed line.

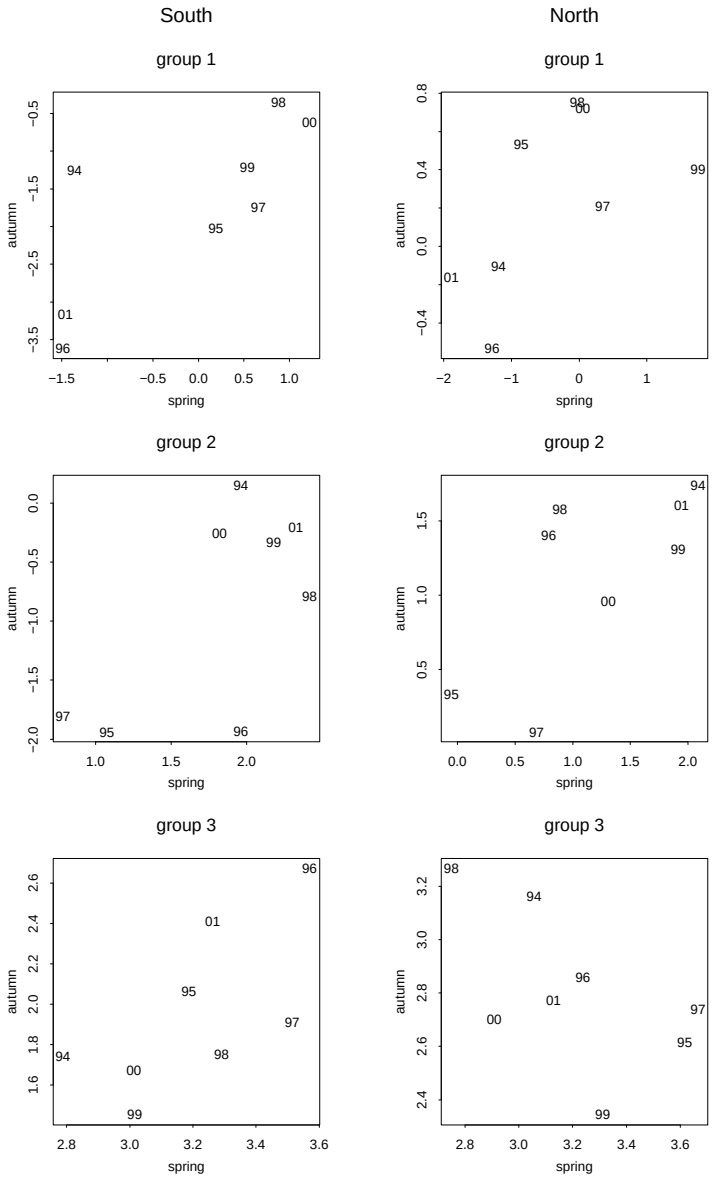


Figure 10: Scatter plots of the log transformed survey indices: autumn survey vs the spring survey for each length group.

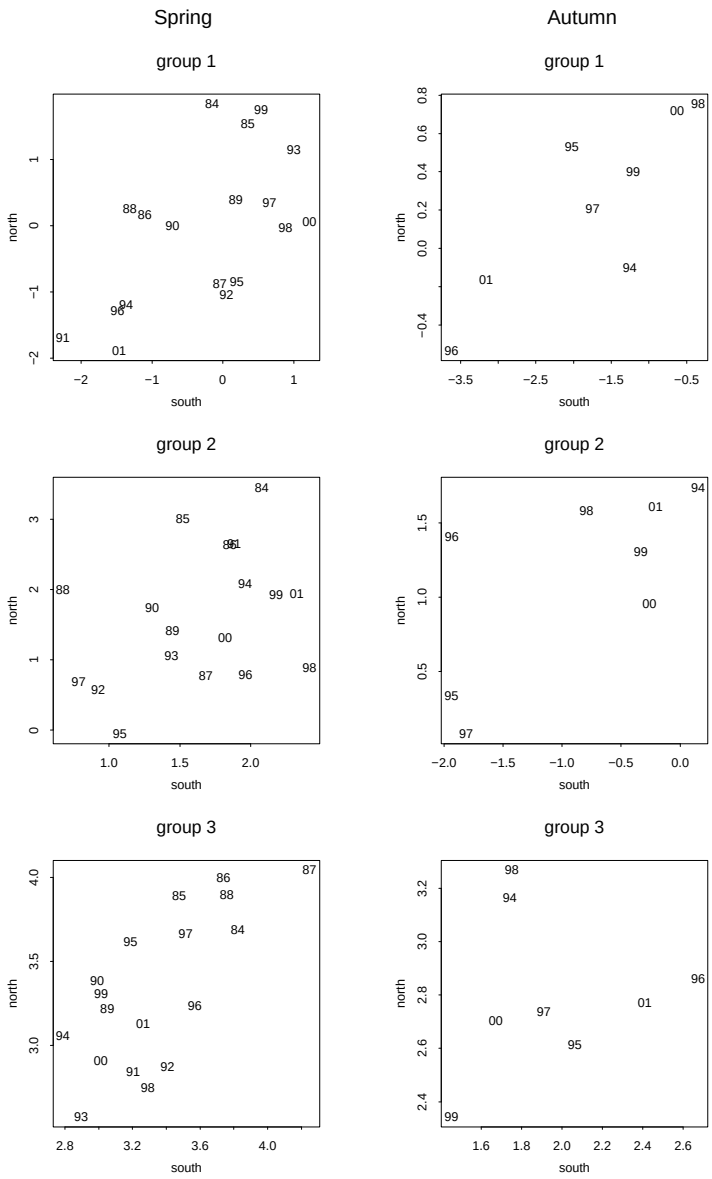


Figure 11: Scatter plots of the log transformed survey indices: for each survey and length group, plots of north vs south.

When the length groups are compared within each survey (figure 12 for the spring survey and figure 13 for the autumn) there are strong correlations between length groups 1 and 2 for both surveys in both areas, indicating little migration of age 1 fish between these areas. The correlations are:

Spring

	South	North
group 1 : group 2	0.705	0.657

Autumn

	South	North
group 1 : group 2	0.670	0.857

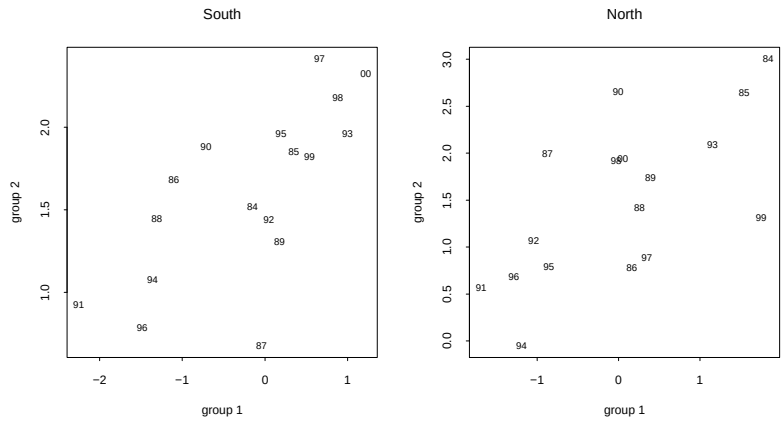


Figure 12: Spring survey: scatter plots of the length group 2 index against the length group 1 index for both areas on a log scale with each point labelled with the cohort year.

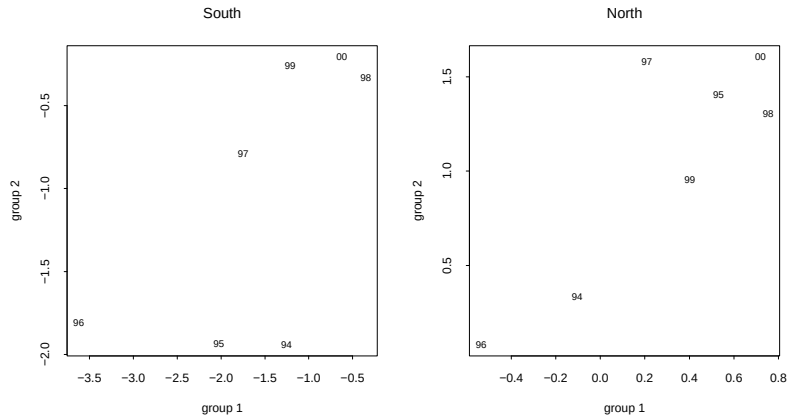


Figure 13: Autumn survey: scatter plots of the length group 2 index against the length group 1 index for both areas of the autumn survey on a log scale with each point labelled with the cohort year.

4.2 Model Implementation

As the model is similar to that described in Taylor et al. (2004) only issues pertaining to multi-area aspects of the model will be described here. Mature cod migrate, with a fixed migration pattern in every year and migration is not age dependent whereas immature cod do not migrate. To reduce computation time, in the development of the 2-area cod model there is no cannibalism. As the main interests of this model are to estimate migration and the distribution of the cod population between the two areas, growth, maturation and fleet suitabilities are fixed from values established from a single area cod model.

Distribution of recruits between the areas is not constant, as indicated in figure 9. To distribute the recruits in every year, either the proportion in each area for each year needs to be fixed or it is necessary to estimate two sets of recruitment parameters.

- The number at age of the initial population (ages 2 to 12) is estimated within the model. Separate parameters describe the initial population on each area i.e. 22 parameters describe the age structure of the initial population.
- The mean and standard deviation of length of the initial population on each area were calculated from survey data and output from the datawarehouse. The weight-length relationships used for weight growth were also used for the initial population in both areas.
- Mean and standard deviation of length of the age 1 recruits was estimated from the March survey for each year, with separate values for each area.
- The number of age 1 recruits for each year (1984–2002) is estimated within the model, with separate recruitment parameters for each area.

- Migration is described by three matrices describing the movement of cod. Matrix 1 represents the spawning migration in January to April, matrix 2 any returns to the north from June to November, and matrix 3 describes no movement (May and December). This results in two migration parameters which need to be estimated.

4.3 Results

With the parameters for growth, selection and maturation fixed, only one optimisation run is required after the weights have been estimated. The weights on the age distribution and survey components are shown in table 2. As the tagging objective function is a Poisson distribution, rather than the sum of squares used for the other components, the appropriate weights for the tagging components cannot be estimated from the standard procedure. In this example the tagging weights were set arbitrarily, but, given the importance of the migration parameters this is an issue which needs to be considered more closely.

There is less difference between the weights of the survey indices for the 2-area model than for the single area case study (Taylor et al. 2004) with most indices of a similar order of magnitude. For the single area model, the weight on length group 2 was the highest for both surveys whereas length group 3 has the highest weight in the south for both surveys and in the north it is only for the autumn survey that length group 2 has a higher weight. This may reflect the relative importance of the north and south for juvenile and mature fish respectively. Alternatively, it may indicate the relative difficulty the model has in fitting the length group 2 indices for the 2-area model.

Component	Weight	sse_f/sse_m
ADs	85.65	1.470
ADc	83.72	1.063
I_{1Ss}	7.63	1.989
I_{2Ss}	12.08	2.631
I_{3Ss}	42.99	4.115
I_{1As}	38.44	6.607
I_{2As}	65.20	4.482
I_{3As}	72.40	8.489
I_{1Sn}	6.00	3.513
I_{2Sn}	22.93	2.651
I_{3Sn}	22.42	2.051
I_{1An}	44.47	1.056
I_{2An}	56.42	1.305
I_{3An}	20.34	1.779

Table 2: Component weights calculated from the minimum sums of squares and sse_f/sse_m, the ratio of the sums of squares from the final model (sse_f) with the minimum sse (sse_m) from the iterative weighting. The components are: AD – age distributions with s and c denoting survey and catch and survey indices I_i, $i = 1, 2, 3$ are the length groups, the S, A subscripts indicate the spring and autumn surveys and the s, n subscripts the southern and northern areas.

The estimated parameters are shown in figure 14 and they indicate that most of the population is in the larger southern area. Migration between the two areas is estimated to be very low with less than 2% of mature fish migrating in each timestep.

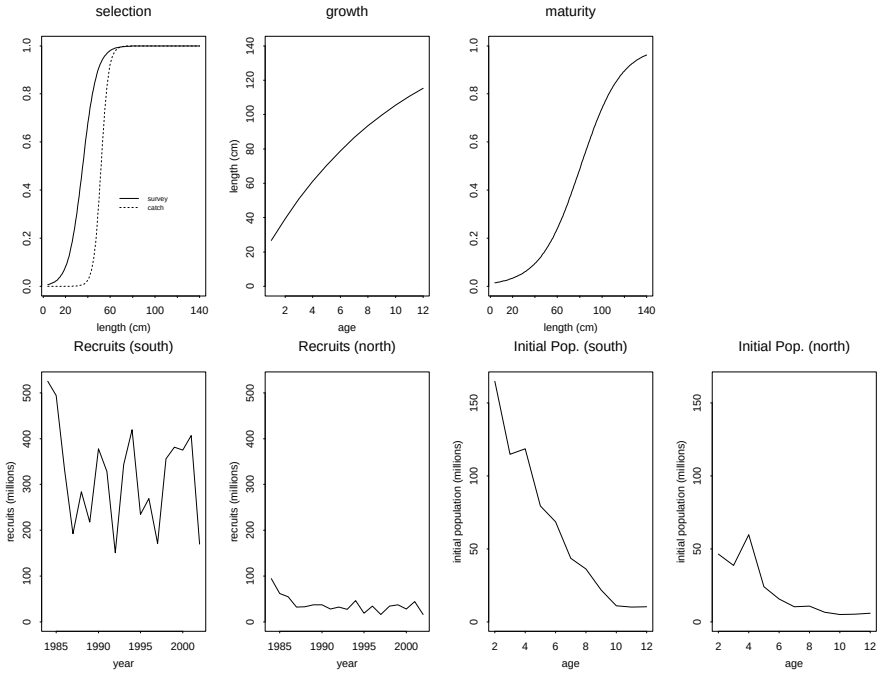


Figure 14: Parameter estimates from the final optimisation: selection patterns of the spring survey and catch (fixed), growth (fixed), maturation (fixed), number of age 1 recruiting into the model for the southern and northern areas and the initial population age 2–12 for the south and north.

The fit of the indices to the model varies between indices (figures 15 & 16). Group 2 in the south is a poor fit for the spring survey and has a slope of 0.67. The equivalent slope for autumn is 3.4 and for the north the slopes are 2.3 and 2.2 for the spring and autumn respectively. In the north, the fit of the model to the indices is worst for length group 3. For both surveys in the north, the slope fitted to the length group 1 indices is less than for the length group 2 indices which is contrary to expectations of catchability of these length groups.

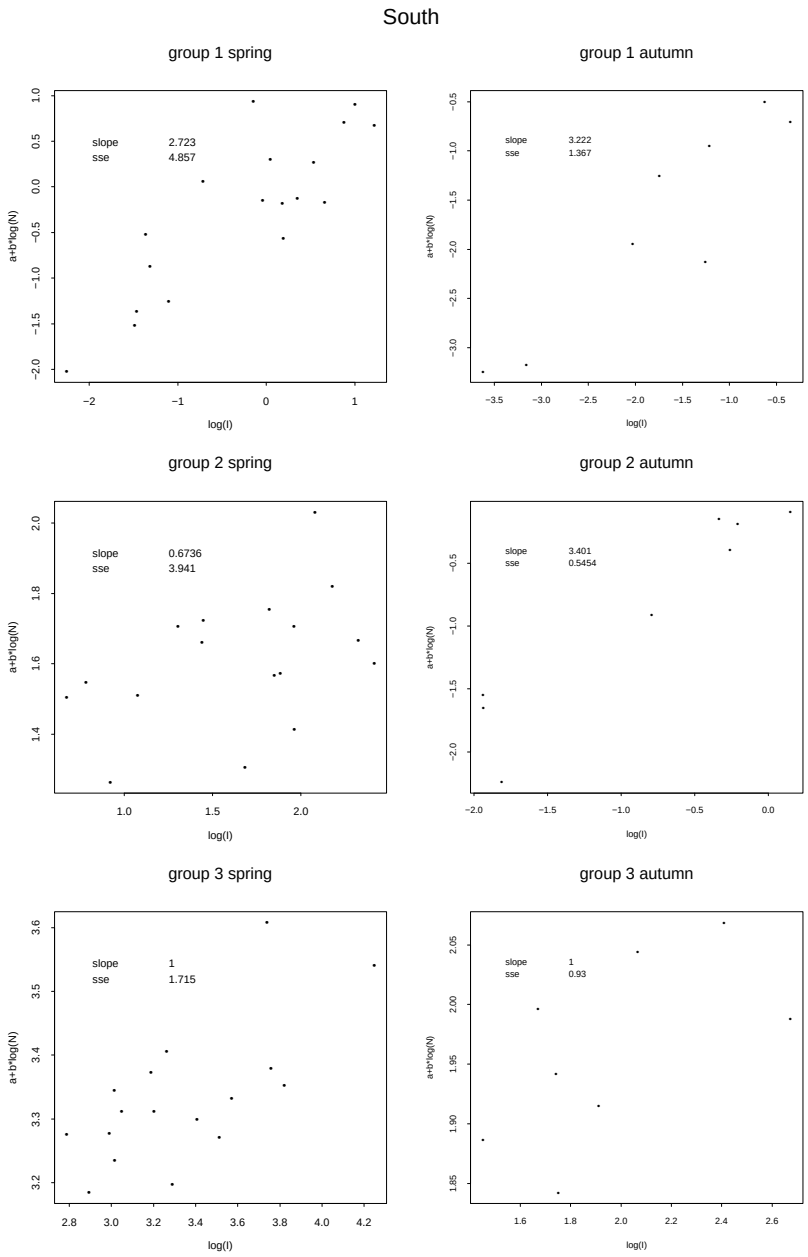


Figure 15: Plots of the log-linear regressions for the southern area as estimated in Gadget, with one plot for each likelihood component and the corresponding slope (estimated for length groups 1 and 2) and sse for each survey index. log(I) is the index and a+b*log(N) the survey index estimated by the model.

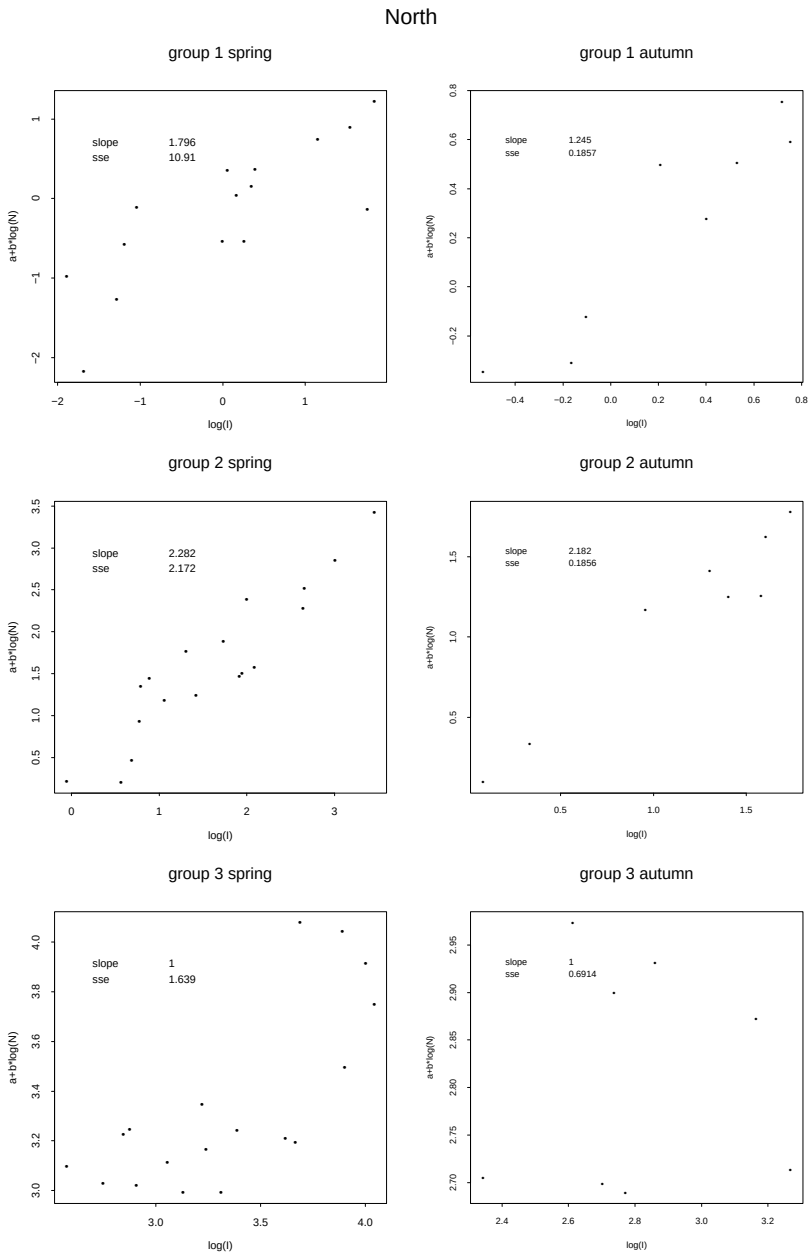


Figure 16: Plots of the log-linear regressions for the northern area as estimated in Gadget, with one plot for each likelihood component and the corresponding slope (estimated for length groups 1 and 2) and sse for each survey index. log(I) is the index and a+b*log(N) the survey index estimated by the model.

The area off the north of Iceland is an important nursery area for cod with the main spawning ground off the south-west coast (Marteinsdottir et al. 2000). This indicates that, at some time, migration must take place from the north to the south, even if only once. The tagging data used as likelihood data were from tagging experiments on the mature stock component, often during the spawning season. If migration, from north to south (given the spatial structure of this model), is primarily of first time spawners migrating from the nursery ground to the southern spawning grounds then these experiments will not provide any information on this pattern of migration. Without migration of this form represented in the model, the abundance of immature cod in the north will be underestimated and overestimated in the south. To include such a migration pattern involves a change in the way the stock is structured in the model and to estimate migration requires tagging experiments on juvenile cod. Tagging data on juvenile cod is limited for recent years but juvenile cod tagged in the north in 2002 have been recaptured off the north-west aged 4 and 5 (Saemundsson, pers. com.). This would represent migration between areas for the spatial structure in this model implementation.

5 Cod-shrimp interaction

The simplest manner in which to to combine the cod and shrimp models is to have cod as predators of shrimp and other food, ignoring cod cannibalism. Cannibalism does not appear to have a large impact on the stock dynamics and with no cannibalism the cod parameters can be fixed to those from a single species 2-area model, reducing the number of parameters estimated. This is a temporary measure for the development of the model and ideally cannibalism would be included and more parameters estimated.

The preference for shrimp by cod, in Icelandic waters, decreases as predator length increases (Magnusson and Palsson 1991b) and as can be seen in figure 17. Magnusson and Palsson (1991b) also found that for small cod there is an increase in prey length as predator length increases. For some areas, predation on male shrimp is considered to exceed that on females as predation is largely by young cod Fu et al. (2001) and data from the north of Iceland indicate that most consumption of shrimp by small cod is on shrimp which are probably male.

Data indicate that shrimp constitute approximately 10% of the biomass in cod stomachs (figure 17) which is the same value found by Magnusson and Palsson (1991b). For both Icelandic waters (Magnusson and Palsson 1991b) and the Flemish Cap (Parsons et al. 1998) no significant relationship the stock biomass/CPUE of the shrimp fishery and the level of predation by cod.

In the stock production model of Stefansson et al. (1994) the consumption index was based on the abundance of the immature part of the cod stock. Two models are considered here, one with predation by all cod and the other with predation only by the immature stock component.

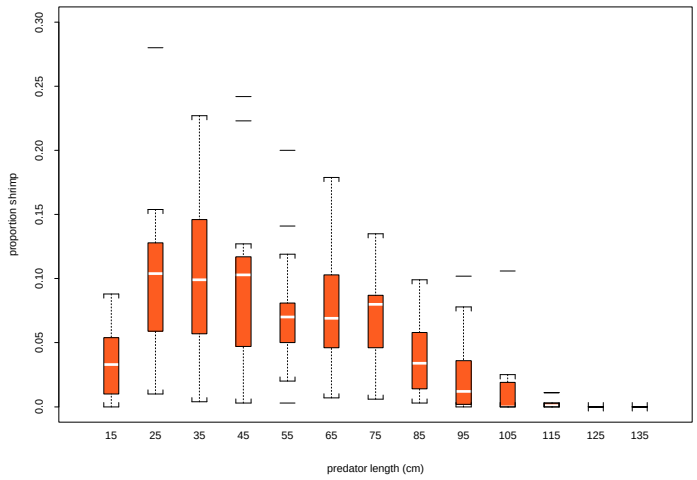


Figure 17: Box plots of the proportion of shrimp in cod stomachs by predator length from the Spring groundfish survey. The stomach contents are aggregated by year and area.

5.1 Data and likelihood functions

The only data required in addition to those described previously are stomach content likelihood data:

- Composition of stomach contents as ratios of prey biomass by species (i.e. shrimp or other) aggregated over all predator lengths from the March groundfish survey for 1988–2002 and other data sources where available. These are combined into one likelihood component.
- Length distribution of prey by predator length groups as biomass ratios from the March groundfish survey for 1988–2002. Predators are aggregated into 20cm length groups and prey into 4mm length groups.
- Length distribution of prey by predator length groups as biomass ratios from other stomach sampling projects. Predators are aggregated into 20cm length groups and prey into 4mm length groups.

5.2 Model Implementation

The model runs from 1988 to 2002, with monthly timesteps, on two areas. There are two species, each with two stock components: immature and mature cod and male and female shrimp. Cod are present on both areas, with shrimp only on one area. The stocks are structured as for the single species models described earlier. There are four fleets, with two operating on each species and no fleet affects both species. The implementation of consumption is described in Taylor et al. (2004) for cod cannibalism and the approach is similar for the two species model.

Two models are considered, one with predation by both immature and mature cod and the other with predation only by immature cod. With predation by all cod a single set of prey suitability parameters is estimated.

5.3 Results

The prey selection patterns estimated within the models are shown in figure 18. The optimal predator length:prey length ratio was fixed to the same value (4) in both cases. Given the observed differences in the size preferences of juvenile and older cod the use of different selection parameters for immature and mature cod should be investigated. This would allow for as increasing size preference for smaller cod but also allow for larger cod being able to consume all length groups.

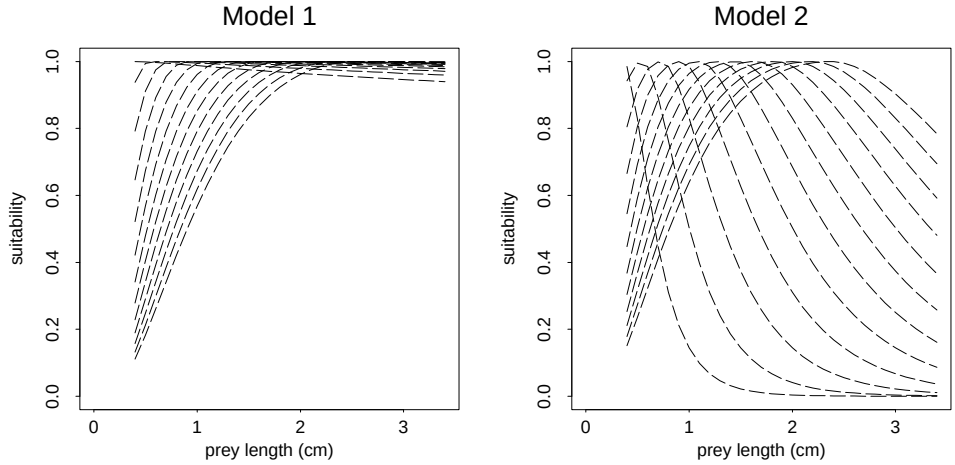


Figure 18: Estimated prey selection patterns with all cod as predators of shrimp (model 1) and with only immature cod as predators (model 2).

The annual consumption of shrimp greatly depends on the model specification as can be seen in figure 19. With predation only by immature cod, annual predation on shrimp by cod is less than the catch, whereas with the additional predation by mature cod, predation exceeds the catch in most years.

Stefansson et al. (1994) estimated cod consumption of the fishable shrimp stock to be 11 600 tonnes (approximately half that estimated here for immature alone), but only shrimp longer than 12mm and consumption by immature cod were considered. Magnusson and Palsson (1991b) estimated the annual consumption of shrimp by cod in Icelandic waters to be around 100 thousand tonnes but this estimate was for all *P. borealis* around Iceland and only the offshore stock is considered in this model.

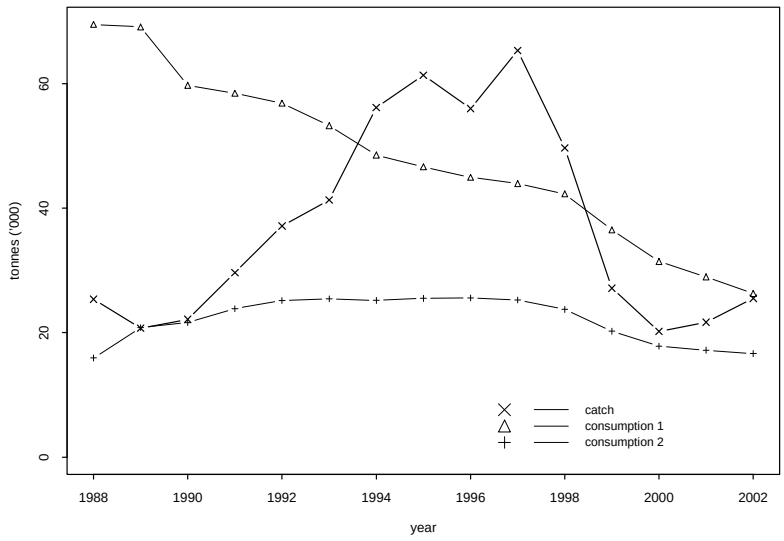


Figure 19: Catch and consumption of shrimp estimated from model 1 (consumption 1) and model 2 (consumption 2).

The fit of both models to the shrimp survey indices is similar (figures 20 & 21) especially for the length group 2 indices. The stock biomass is very large in both cases and as with the single species model increases with time as the number of older shrimp continuously increases. The high biomass means that predation has little impact on the stock structure and dynamics.

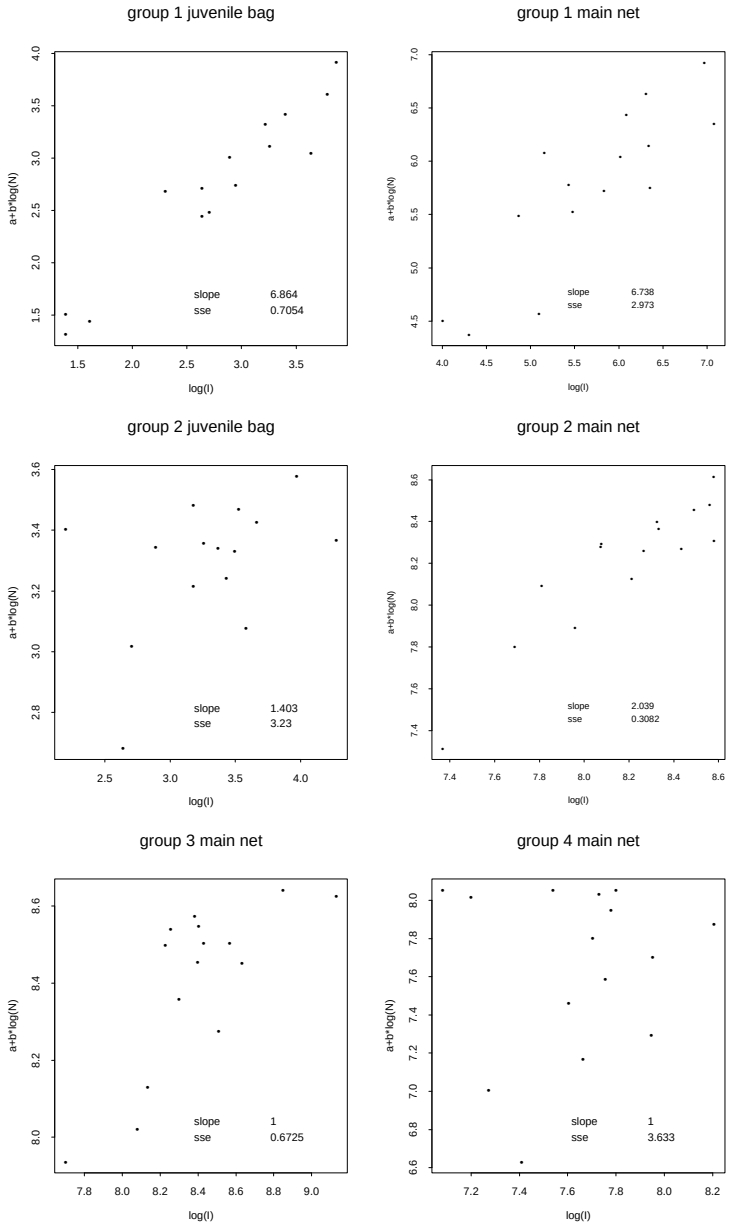


Figure 20: Shrimp survey indices from model with predation from immature and mature cod: plots of the log-linear regression as estimated in Gadget, with one plot for each likelihood component and the corresponding slope (estimated for length groups 1 and 2) and sse for each survey index. log(I) is the index and a+b*log(N) the model estimate of the survey index.

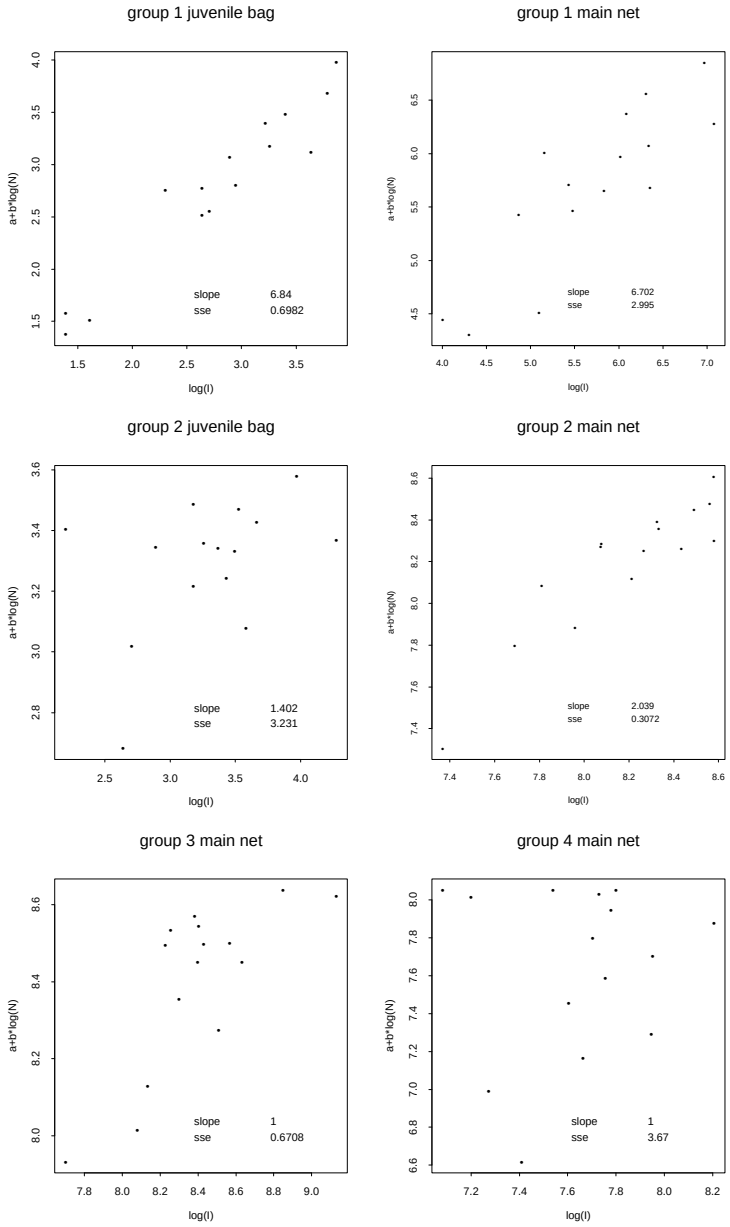


Figure 21: Shrimp survey indices from model with predation only by immature cod: plots of the log-linear regression as estimated in Gadget, with one plot for each likelihood component and the corresponding slope (estimated for length groups 1 and 2) and sse for each survey index. log(I) is the index and a+b*log(N) the model estimate of the survey index.

Discussion

Shrimp: There is clearly a problem in estimating the total biomass of the shrimp population and in determining the age structure. This is a common problem for shrimp due to the lack of age data and variable natural mortality. Given the ability of the model to fit the survey indices for the younger age groups, the mortality of the older shrimp, especially that of mature females, needs to be considered more closely. It would also be useful to find a way of creating more than one index for the older shrimp in order to be able to determine more appropriate weights for these likelihood components.

Cod: In a 2-area model, age data are required along with the survey to estimate the total population size as migration breaks the direct link between the oldest survey group (which in a single area model is an important determinant of the level of abundance) and recruitment. This means, however, that is it probably necessary to estimate the initial population abundance, recruitment and the migration parameters simultaneously. The possible importance of including more than one commercial fleet is discussed for the single area cod model in Taylor et al. (2004) and with more than one area, this would be expected to become a more important issue as the distribution of gear use on the different areas is not homogeneous.

To more accurately determine the distribution of cod between the areas, it is necessary to consider the migration of juveniles from the north to the south. This could be incorporated into the current implementation by including an additional cod component. It may also be useful to fix migration to externally estimated parameters and compare with tagging data rather than estimate migration directly from relatively limited mark-recapture data from juvenile cod. These areas are not of limited interest in terms of migration patterns of mature cod but migration of juveniles is expected to be significant.

Cod and shrimp: The predation of shrimp by cod needs to be considered in more detail. The decreasing preference for shrimp by cod should be incorporated into the model and the size preferences analysed more carefully. Length distributions from stomach contents indicate low consumption of shrimp of lengths 18–22mm for all but the largest cod. This is most probably due to spatial heterogeneity of the shrimp stock and shrimp are known to migrate in other area (e.g. the Gulf of Maine (Clark et al. 2000)). These bimodal length distributions, are expected to cause problems in parameter estimation. Given the problems in estimating migration the benefits of using more areas should be evaluated.

Predation of shrimp by cod is determined more by the abundance of cod than availability of shrimp and to accurately determine consumption the cod stock abundance and structure in the northern area needs to be modelled accurately. For any information on the impact of predation on the shrimp stock dynamics, more certainty is required on the biomass of the shrimp stock.

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